TH-DAT: Trimester-Aware Hierarchical Domain-Attention Transformer for Antenatal Depression Prediction

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Abstract

Antenatal depression remains a critically underdiagnosed condition affecting 10-20% of pregnant women worldwide, contributing to adverse maternal and neonatal outcomes including preterm birth, low birth weight, impaired maternal-infant bonding, and elevated postnatal depression risk. Current machine learning approaches treat clinical features as flat inputs, ignoring clinically meaningful domain groupings and pregnancy-stage variation across trimesters. We propose TH-DAT (Trimester-Aware Hierarchical Domain-Attention Transformer), a novel deep learning architecture introducing: (1) domain-grouped feature tokenization with domain-aware positional encoding, (2) a global transformer encoder enabling cross-domain feature interactions, (3) trimester-sequential cross-attention conditioning predictions on pregnancy stage, and (4) gated domain fusion with entropy regularization providing patient-level interpretability. TH-DAT employs two-phase training (masked pretraining followed by focal-contrastive fine-tuning) with SMOTE oversampling. Evaluated on PERI_DEP (N=14,008, Pakistan) and Uganda EPDS (N=14,325), TH-DAT achieved an AUC-ROC of 0.9440 on the PERI_DEP dataset and 0.9026 on the Uganda EPDS dataset, outperforming all 11 baselines including Random Forest (0.9369), XGBoost (0.8529), TabTransformer (0.8971), and text transformers (BERT, GPT-2, RoBERTa, T5). A six-configuration ablation study quantified the performance changes associated with removing individual architectural components. SHAP analysis showed broadly consistent domain-level feature attribution with the learned gate weights (Demographic: 31.6%, Obstetric: 34.1%, Psychosocial: 31.6%).

Keywords: Antenatal depression, transformer, tabular data, attention mechanism, clinical decision support, interpretable AI, SMOTE, SHAP, domain attention, multi-cohort validation

1. Introduction

Depression during pregnancy, termed antenatal depression (AD), represents the most prevalent psychiatric complication of the perinatal period, affecting an estimated 10 to 20 percent of women globally. The World Health Organization recognizes maternal mental health disorders as a leading cause of disease burden among women of reproductive age, with disproportionate impact in low- and middle-income countries

(LMICs) where up to 25% of pregnant women experience clinically significant depressive symptoms. Untreated antenatal depression carries significant consequences for both mother and offspring, including elevated risk of preterm birth (OR 1.37), intrauterine growth restriction, low birth weight (OR 1.28), compromised maternal-infant bonding, inadequate prenatal care utilization, substance use during pregnancy, and a 3 to 5 fold increased probability of developing postnatal depression.

Despite its clinical significance and well-documented adverse outcomes, systematic screening for antenatal depression remains inconsistent across healthcare systems globally. Standard screening protocols rely on validated self-report instruments such as the Edinburgh Postnatal Depression Scale (EPDS), developed by Cox et al. (1987), and the Patient Health Questionnaire-9 (PHQ-9), validated by Kroenke et al. (2001). These instruments require trained personnel for administration and interpretation. In resource-constrained settings, where the burden of antenatal depression is paradoxically highest, routine screening is frequently omitted due to workforce limitations, competing clinical priorities during prenatal visits, stigma surrounding mental health, and lack of mental health infrastructure and referral pathways.

An automated predictive system that leverages routinely collected clinical data, including demographic background, obstetric history, and psychosocial indicators, could enable proactive identification of at-risk women during standard antenatal care visits, without requiring dedicated screening instruments or specialized mental health training. Such a system must satisfy three critical requirements: (i) high predictive accuracy to minimize both missed cases (false negatives that deny care to at-risk women) and unnecessary referrals (false positives that burden limited mental health services), (ii) clinical interpretability enabling practitioners to understand and act upon predictions in terms of clinically meaningful factors, and (iii) cross-population robustness ensuring reliable performance across diverse healthcare contexts, cultural settings, and socioeconomic conditions.

1.1 Research Challenges

C1. Domain Structure Ignorance: Clinical features naturally cluster into demographic (age, education, employment), obstetric (gestational age, parity, complications), and psychosocial (physical health perception, body image, family relationships) domains. Existing ML models treat these as flat, exchangeable input vectors, discarding the intra-domain correlations (e.g., education correlates with employment within demographics) and inter-domain interactions (e.g., obstetric complications exacerbate psychosocial stress) that are critical for accurate risk assessment.

C2. Temporal Insensitivity: Depression risk profiles evolve across pregnancy trimesters. First-trimester risk factors (unplanned pregnancy, morning sickness, adjustment difficulties) differ substantially from third-trimester factors (delivery anxiety, financial preparation stress, physical discomfort). To our knowledge, no widely adopted computational model explicitly captures this temporal conditioning of pregnancy-related depression risk, treating trimester as just another categorical input feature.

C3. Class Imbalance: Depression prevalence in antenatal populations typically ranges from 15-35%, creating significant class imbalance. Models trained with standard cross-entropy loss achieve deceptively high overall accuracy by preferentially predicting the majority class (non-depressed), resulting in poor sensitivity for the minority class; precisely the patients who need clinical attention are systematically missed by such models.

C4. Interpretability Deficit: Clinical adoption of AI-based screening tools requires transparent, actionable predictions. Clinicians need to understand why a patient is flagged for depression risk in terms of clinically meaningful domain-level factors, not merely receive an opaque probability score. Post-hoc methods such as SHAP and LIME provide explanations but are computationally expensive and may produce inconsistent explanations across repeated runs, undermining clinical trust.

C5. Cross-Cohort Robustness Gap: Models developed and validated on single datasets from specific populations may not transfer reliably across cohorts with different cultural, socioeconomic, and healthcare contexts. Cross-cohort evaluation using geographically and culturally diverse datasets is essential for establishing clinical credibility yet is rarely performed in perinatal depression prediction research.

1.2 Contributions

This paper makes the following seven contributions:

- **Novel Architecture:** TH-DAT introduces a transformer architecture specifically designed for antenatal clinical tabular data, integrating domain-grouped feature tokenization, trimester conditioning, and gated domain fusion.
- **Comparative Performance Evaluation:** TH-DAT achieves the highest AUC-ROC among the evaluated models of 0.9440 among all 12 models evaluated on the PERI_DEP dataset, outperforming Random Forest (0.9369, +0.76%), TabTransformer (0.8971, +5.23%), XGBoost (0.8529, +10.68%), and BERT (0.6689, +41.12%). On Uganda EPDS, TH-DAT achieves 0.9026 AUC-ROC.
- **Two-Phase Training with SMOTE:** Combined masked-feature pretraining and focal-contrastive fine-tuning with SMOTE oversampling, simultaneously addressing representation learning, loss sensitivity, and data-level class imbalance in a unified training framework.
- **Comprehensive Benchmark:** Systematic comparison of 12 models across four categories (classical ML, deep learning, tabular transformers, text transformers) on two international datasets comprising 28,333 total records, representing a broad comparative evaluation in perinatal depression prediction.
- **Systematic Ablation:** Six-configuration ablation study quantifying each architectural component contribution, with skip connection (-1.96%) and trimester attention (-1.47%) identified as the two most impactful components.
- **Clinical Interpretability:** Per-patient domain gate weights provide model-derived domain-level attribution signals, which showed broad agreement with SHAP analysis (mean absolute discrepancy of 0.53 percentage points across three clinical domains on the held-out test set; see Table 17).

- **Multi-Cohort Validation:** Consistent performance across Pakistani (PERI_DEP) and Ugandan (EPDS) populations with different cultural contexts, healthcare systems, and depression prevalence rates, providing preliminary evidence of transportability across the two evaluated cohorts.

2. Literature Survey

The application of machine learning and deep learning to perinatal depression prediction has grown substantially in recent years. This section provides a structured review of 18 representative recent studies identified through PubMed, Google Scholar, and IEEE Xplore searches using terms including antenatal depression prediction, perinatal depression machine learning, and prenatal mental health deep learning (2019-2026) organized by methodology, followed by an analysis of existing tabular transformer architectures and identification of the research gap addressed by TH-DAT. Table 1 provides a systematic comparison of all reviewed studies.

2.1 Classical Machine Learning Approaches

Krishnamurti et al. (2024) applied classical machine learning classifiers to patient-reported pregnancy data for predicting first-time depression onset during pregnancy, achieving an AUC of 0.89. Their work demonstrated the predictive value of self-reported clinical information but relied on traditional flat-feature representation without domain-specific feature processing or temporal conditioning. Huang et al. (2024) specifically addressed the critical challenge of model bias in prenatal depression prediction using ML with systematic bias assessment techniques, achieving only 0.61 AUC, highlighting the fundamental tension between ensuring demographic fairness and maintaining predictive accuracy in clinical ML systems.

Hu et al. (2025) developed an ML-based antenatal depression prediction model using clinical maternal records, achieving 0.710 AUC. Their moderate performance underscores the difficulty of predicting depression from routine clinical data without specialized feature engineering. Chen et al. (2025) combined elastic net regression with ML approaches, leveraging both clinical and biochemical markers including remnant cholesterol for peripartum depression prediction (0.875 AUC), demonstrating the potential of multimodal feature integration beyond standard questionnaire-based approaches.

2.2 Multiple Model Comparison Studies

Qi et al. (2025) systematically developed and validated multiple ML models for postpartum depression prediction across different clinical settings, reaching 0.909 AUC with their best-performing model. Wang et al. (2025) pursued interpretable ML for predicting perinatal depression specifically in Chinese women during mid- and late pregnancy, achieving 0.917 AUC with both development and external validation cohorts, representing one of the highest reported performances. Han et al. (2023) developed a comprehensive psychological tendency prediction model based on standardized questionnaire responses, reporting 92.3% accuracy using ensemble methods.

2.3 Ensemble Methods and Maternal Health Risk

Khadidos et al. (2024) proposed an ensemble ML framework for assessing maternal health risk during pregnancy, achieving 98.4% accuracy, though their scope addressed general health risk classification rather than depression-specific prediction. Pi et al. (2025) applied ML algorithms to predict high-risk pregnancy complications with 94.6% accuracy. While both demonstrate the strong potential of ensemble approaches for maternal health classification, neither incorporates domain-aware feature processing or temporal pregnancy stage modeling.

2.4 Novel Data Modalities and Multimodal Approaches

Wiley et al. (2025) explored voice biomarkers for perinatal depression detection, achieving 0.78 AUC, demonstrating that paralinguistic features in speech carry clinically meaningful depression signals. Zhang et al. (2024) combined LightGBM with conversational speech features for depression screening in naturalistic conditions (0.79 AUC). These studies represent promising future directions for multimodal depression assessment that could complement tabular clinical data.

2.5 Risk Factor Identification and Explainable Models

Sibbald et al. (2025) used LASSO-regularized ML incorporating prenatal psychological, biological, and social data to identify key postpartum depression risk factors (0.72 AUC). Lin and Zhou (2025) proposed an ensemble neural network for postpartum depression (90.0%). Huang et al. (2025) applied explainable ML techniques (0.91 AUC). Zhang et al. (2025) developed interpretable ML models (0.89 AUC). Kim (2025) conducted predictive analysis of postpartum depression (0.88 AUC). Wang et al. (2024) studied longitudinal postpartum depression trajectory patterns (0.84 AUC). Malani et al. (2026) predicted postpartum psychiatric conditions from a large Queensland cohort of 205,495 pregnancies using gradient-boosted trees (0.79 AUC), representing the largest-scale study in this domain.

2.6 Transformer Architectures for Tabular Data

Huang et al. (2020) introduced TabTransformer, the first application of multi-head self-attention to tabular data, applying transformers exclusively to categorical features while processing numerical features through a separate pathway. Gorishniy et al. (2021) proposed FT-Transformer, which tokenizes all features (both categorical and numerical) with a [CLS] token for classification, representing the first fully attention-based tabular model. Somepalli et al. (2021) introduced SAINT with a novel intersample attention mechanism alongside standard self-attention. Despite these architectural advances, all existing tabular transformers lack three critical capabilities for clinical depression prediction: domain-specific feature processing, temporal pregnancy stage conditioning, and built-in interpretability through learned domain contribution weights.

2.7 Research Gap Analysis

Across all 18 reviewed studies and existing tabular transformer architectures, to the best of our knowledge based on the literature reviewed in this study, no prior antenatal-depression prediction framework simultaneously integrates these five components: (1) domain-aware feature processing that respects clinical groupings, (2) trimester-based conditioning that captures evolving risk profiles, (3) gated interpretable fusion with entropy regularization for patient-level explanations, (4) self-supervised masked pretraining for tabular clinical data that learns feature correlations without labels, and (5) multi-cohort validation across geographically and culturally diverse populations. TH-DAT addresses all five requirements in a unified, end-to-end trainable framework.

Table 1. Comprehensive Comparison of ML-Based Studies for Antenatal and Perinatal Depression Prediction

S.No.	Authors (Year)	Method	Dataset	Best Result

1	Krishnamurti (2024)	ML Classifiers	Patient-reported pregnancy	0.89 AUC
2	Huang et al. (2024)	ML + Bias Assess.	Prenatal clinical records	0.61 AUC
3	Hu et al. (2025)	ML Prediction	Clinical maternal records	0.710 AUC
4	Chen et al. (2025)	Elastic Net + ML	Clinical + biochemical	0.875 AUC
5	Qi et al. (2025)	Multiple ML	Maternal clinical data	0.909 AUC
6	Wang et al. (2025)	Explainable ML	Chinese maternal cohort	0.917 AUC
7	Han et al. (2023)	Psychological Pred.	Questionnaire responses	92.3%
8	Khadidos (2024)	Ensemble ML	Maternal health risk	98.4%
9	Pi et al. (2025)	ML Algorithms	High-risk pregnancy	94.6%
10	Wiley et al. (2025)	Voice Biomarker	Speech recordings	0.78 AUC
11	Sibbald et al. (2025)	LASSO-Reg. ML	Prenatal bio/social	0.72 AUC
12	Lin & Zhou (2025)	Ensemble NN	Postpartum depression	90.0%
13	Huang et al. (2025)	Explainable ML	Clin./demog. maternal	0.91 AUC
14	Zhang et al. (2025)	Interpretable ML	Retrospective clinical	0.89 AUC
15	Kim (2025)	Machine Learning	Postpartum clinical	0.88 AUC
16	Zhang et al. (2024)	LightGBM+Speech	Conversational speech	0.79 AUC
17	Wang et al. (2024)	Machine Learning	Longitudinal postpartum	0.84 AUC
18	Malani et al. (2026)	Grad-Boost Trees	Queensland (205K)	0.79 AUC
19	**TH-DAT (Ours)**	**Domain-Aware TF**	**PERI_DEP + Uganda**	**0.944 AUC**

3. Problem Formulation

3.1 System Model and Notation

Let $D = \{(x_i, t_i, y_i)\}$ for $i = 1$ to M denote a clinical dataset of M antenatal patient records, where x_i in R^N is the N-dimensional clinical feature vector, t_i in $\{0, 1, 2\}$ is the trimester indicator derived from gestational age, and y_i in $\{0, 1\}$ is the binary depression label (0 = non-depressed, 1 = depressed). The objective is to learn a parameterized function f_θ: $R^N \times \{0,1,2\} \rightarrow [0,1]$ that accurately predicts depression probability from clinical features and trimester information.

3.2 Domain Partitioning

The N feature indices are partitioned into three clinically defined, non-overlapping domain groups: I_{demo} (demographic features), I_{obst} (obstetric features), and I_{psyc} (psychosocial features), such that $I_{demo} \cup I_{obst} \cup I_{psyc} = \{1, ..., N\}$ and all pairwise intersections are empty. For the PERI_DEP dataset (N=19): $|I_{demo}| = 6$, $|I_{obst}| = 10$, $|I_{psyc}| = 3$. For the Uganda EPDS dataset (N=13): $|I_{demo}| = 4$, $|I_{obst}| = 5$, $|I_{psyc}| = 4$. This partitioning is specified a priori based on clinical domain expertise.

3.3 Trimester Derivation

The trimester indicator t is derived from gestational age g (in weeks): $t = 0$ if g is at most 13 weeks (first trimester, T1), $t = 1$ if g is between 14 and 26 weeks (second trimester, T2), $t = 2$ if g is 27 weeks or greater (third trimester, T3). For the Uganda dataset where gestational age is not directly available, it is computed from: $g = 40 - (\text{due\_date} - \text{evaluation\_date}) / 7$ days.

3.4 Composite Loss Function

$$L\_total = L\_focal + 0.03 * L\_contrastive + 0.01 * L\_entropy$$

where L_focal is focal loss with gamma = 2.0 (Lin et al. 2017), down-weighting easy-to-classify examples to focus training on hard boundary cases; L_contrastive is supervised contrastive loss with temperature tau = 0.1 (Khosla et al. 2020), pulling same-class representations together and pushing different-class representations apart in the embedding space; and L_entropy is entropy regularization on the batch-averaged gate weights to prevent domain mode collapse.

4. Proposed TH-DAT Architecture

4.0 Architectural Overview

TH-DAT processes clinical tabular data through five hierarchical, progressively abstracting components (Figure 1), parameterized by approximately 1,693,220 trainable parameters. The architecture systematically builds patient-level representations from individual scalar feature tokens through domain-level summaries to a final prediction with interpretable per-patient domain contribution weights. Figure 1 presents the complete architectural diagram.

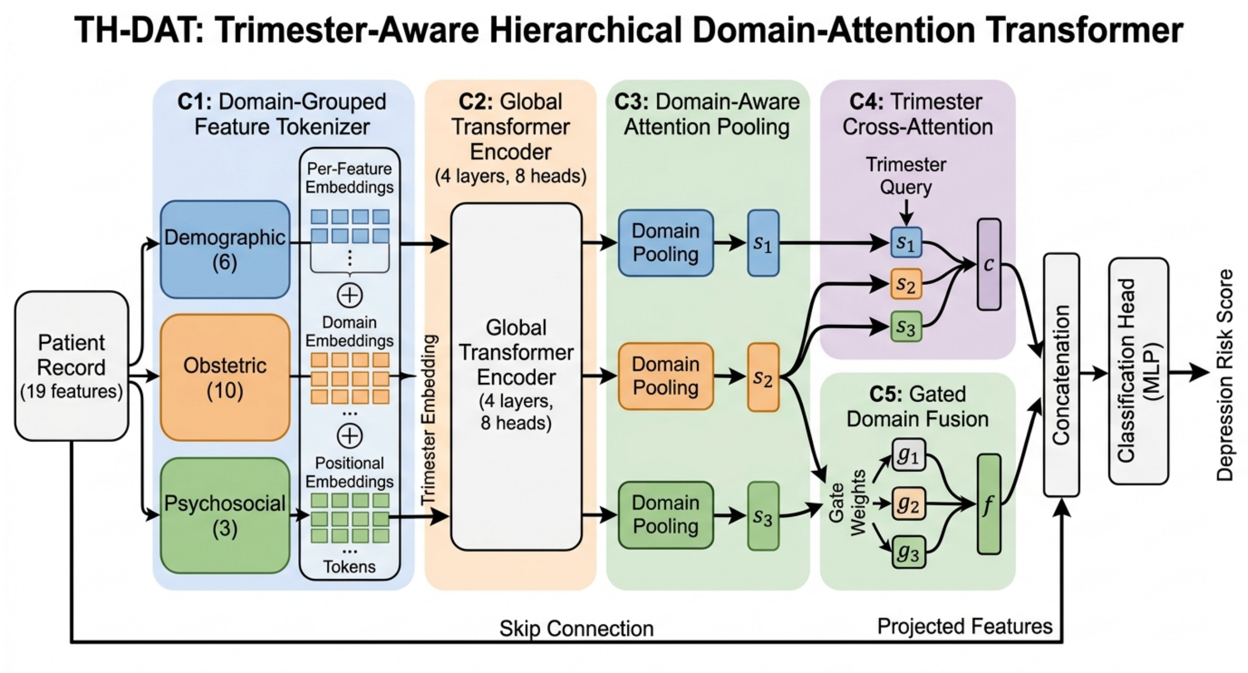

Figure 1.

Figure 1. Complete TH-DAT architecture. Patient records (19 features) are partitioned into three clinical domains: Demographic (blue, 6 features), Obstetric (orange, 10 features), Psychosocial (green, 3 features). Data flows through: C1 Domain-Grouped Feature Tokenizer, C2 Global Transformer Encoder with Trimester Conditioning (L=6, H=8), C3 Domain-Aware Attention Pooling (3 learnable queries), C4 Trimester Cross-Attention, C5 Gated Domain Fusion with entropy regularization. A skip connection from raw input preserves direct feature information for the final classifier.

4.1 Component 1: Domain-Grouped Feature Tokenizer

Each scalar clinical feature x_i is individually projected into a d-dimensional embedding space (d=128) through a feature-specific linear transformation followed by nonlinear activation and normalization:

$$e_i = \mathrm{LayerNorm}(\mathrm{GELU}(W_i * x_i + b_i)), \quad W_i \in R^{(d \times 1)}$$

Unlike TabTransformer and FT-Transformer which share embedding projection weights across features, TH-DAT employs N separate linear projection layers (one per feature), allowing each clinical measurement to learn its own optimal embedding function. The token sequence is then augmented with dual positional information:

$$h_i = e_i + p_i + d_{D(i)}$$

where p_i in R^d is a learnable positional embedding encoding feature identity within the sequence, and $d_{D(i)}$ in R^d is a domain embedding that maps feature i to its clinical domain (demographic, obstetric, or psychosocial). The domain embedding enables the subsequent transformer layers to distinguish features from different clinical domains, a capability absent from all prior tabular transformers.

4.2 Component 2: Global Transformer Encoder with Trimester Conditioning

Before entering the transformer encoder, trimester information is injected as a scaled additive bias to each feature token:

$$h'_i = h_i + \alpha * \mathrm{Emb}_{tri}(t), \quad \alpha = 0.1$$

The scaling factor alpha = 0.1 prevents trimester conditioning from dominating the feature-level information while still enabling the transformer to learn trimester-dependent feature interactions. The augmented token sequence H = [h'_1, ..., h'_N] is processed by a standard transformer encoder with L = 6 layers, H = 8 attention heads, d_{model} = 128, d_{ff} = 512 feed-forward dimension. Each layer applies multi-head self-attention (MSA) followed by a position-wise feed-forward network (FFN) with pre-norm residual connections and dropout (p = 0.1). The global self-attention mechanism enables both intra-domain interactions (e.g., age-education correlations within demographics) and cross-domain interactions (e.g., gestational age-physical health interactions across obstetric and psychosocial domains) to be learned simultaneously.

4.3 Component 3: Domain-Aware Attention Pooling

For each clinical domain k in {1, 2, 3} (demographic, obstetric, psychosocial), a learnable query vector q_k in $R^{(1 \times d)}$ attends exclusively over the transformed tokens belonging to that domain:

$$s_k = \mathrm{LayerNorm}(q_k + \mathrm{MHA}(q_k, H_{Ik}, H_{Ik}))$$

This produces a domain summary matrix S = [s_1; s_2; s_3] in $R^{(3 \times d)}$, where each row captures the essential information from one clinical domain as enriched by the global transformer's cross-domain attention patterns. The attention pooling operation is more expressive than simple mean or max pooling, as it learns which features within each domain are most informative for depression prediction.

4.4 Component 4: Trimester Cross-Attention

The trimester embedding serves as a query attending over the three domain summaries to produce a trimester-conditioned patient representation:

$$c_0 = \mathrm{LayerNorm}(t_q + \mathrm{MHA}(t_q, S, S))$$

$$c = \mathrm{LayerNorm}(c_0 + \mathrm{Dropout}(\mathrm{FFN}_{tri}(c_0)))$$

This cross-attention mechanism allows the model to learn trimester-specific attention patterns over clinical domains. For example, the model can learn that psychosocial factors dominate first-trimester risk while obstetric complications become more predictive in the third trimester, capturing the evolving nature of antenatal depression risk across pregnancy.

4.5 Component 5: Gated Domain Fusion with Entropy Regularization

A two-layer MLP with softmax activation produces per-patient domain contribution weights:

$$g = \mathrm{Softmax}(W_2 * \mathrm{GELU}(W_1 * [s_1; s_2; s_3] + b_1) + b_2)$$

The gate vector $g = [g_1, g_2, g_3]$ satisfies $\mathrm{sum}(g_k) = 1$ and $g_k >= 0$ by construction of the softmax function. The gate layer weights are initialized to zeros, ensuring balanced starting weights of approximately (1/3, 1/3, 1/3). The gated domain representation is computed as the weighted sum:

$$f = g_1 * s_1 + g_2 * s_2 + g_3 * s_3$$

Entropy regularization prevents the gate from collapsing to attend to a single domain:

$$L_{ent} = -\mathrm{SUM}(\bar{g}_k * \log(\bar{g}_k + \epsilon))$$

where $\bar{g}_k$ is the batch-averaged gate weight for domain k and ϵ = 1e-8 prevents numerical instability.

4.6 Final Classification with Skip Connection

Three complementary representations are concatenated for the final classifier:

$$z = [c \,;\, f \,;\, \mathrm{GELU}(W_{skip} * x)] \quad \text{in} \quad R^{(3d)}$$

where c is the trimester-conditioned representation, f is the gated domain fusion, and $\mathrm{GELU}(W_{skip} * x)$ is the skip connection from the raw input features. A three-layer MLP classifier with BatchNorm, GELU activation, and Dropout (0.3) produces the final sigmoid prediction. The skip connection ensures gradient flow and preserves direct access to raw feature values for the classifier.

4.7 Two-Phase Training Strategy

Phase 1 - Self-Supervised Masked Pretraining (80 epochs):

Random 20% of input features are masked (set to zero) and reconstructed via MSE loss using the transformer encoder. This phase learns inter-feature statistical dependencies (e.g., age-education, gravida-children correlations) before exposure to any depression label information, enabling better feature representations.

Phase 2 - Supervised Fine-tuning (300 epochs):

Combined focal + contrastive + entropy loss (as defined in Section 3.4). AdamW optimizer (lr=5e-4, weight_decay=1e-4), CosineAnnealingWarmRestarts scheduler (T_0=50, T_mult=2), MixUp data augmentation (alpha=0.2, applied with 50% probability), label smoothing (epsilon=0.05), Stochastic Weight Averaging (SWA) after epoch 200, and gradient clipping at norm 1.0.

5. Datasets and Preprocessing

5.1 PERI_DEP Dataset (Primary Evaluation)

The PERI_DEP dataset contains 14,008 antenatal records collected from multiple healthcare facilities across Pakistan, made publicly available on Zenodo (DOI: 10.5281/zenodo.11094957). The original dataset contains 27 columns including demographic information, obstetric history, psychosocial indicators, and PHQ-9 screening responses. After removing the 9 PHQ-9 item columns (to prevent data leakage, since these items constitute the depression screening instrument itself) and engineering the Trimester feature from gestational age, 19 predictive features remain organized into three clinical domains:

- **Demographic:** Age, Female Education, Husband Education, Working Status, Sufficient Money, Family System (6 features)
- **Obstetric:** Gestational Age, Num Sons, Num Daughters, Total Children, Gravida, Miscarriage, Gender Preference, Trimester, and additional obstetric indicators (10 features)
- **Psychosocial:** Physical Health perception, Appearance Acceptance, Mother-in-Law Relationship (3 features)

Depression label: df['Labelling'] == 'Depressed' mapped to 1, otherwise 0. Depression prevalence: 64.8% (9,078 depressed out of 14,008 total records). Unlike many antenatal depression datasets where depression is the minority class, PERI_DEP exhibits a higher prevalence of positive labels. Consequently, SMOTE was applied to the non-depressed class (4,930, 35.2%) during training to obtain a balanced 1:1 training distribution.

5.2 Uganda EPDS Dataset (External Validation)

The Uganda EPDS dataset comprises 14,325 prenatal records from 11 hospitals in Kampala and Wakiso districts (Mendeley Data, DOI: 10.17632/4swmy34scp.3). Depression is defined by EPDS total score >= 13 (standard clinical cutoff), yielding 3,299 depressed cases (23.0% prevalence). The dataset contains 13 predictive features organized across three clinical domains: Demographic (4), Obstetric (5), Psychosocial (4). Gestational age is derived from expected delivery date: g = 40 - (due_date - evaluation_date) / 7.

5.3 SMOTE Oversampling Protocol and Data Pipeline

The Synthetic Minority Over-sampling Technique (SMOTE; Chawla et al. 2002) generates synthetic minority-class samples through k-nearest-neighbor interpolation: x_syn = x_i + lambda * (x_nn - x_i), where lambda is drawn from Uniform(0,1). Our SMOTE protocol uses k=5 nearest neighbors with a 1:1 oversampling ratio achieving full class balance. Trimester recovery is performed via quantile binning on interpolated gestational age values.

Data Processing Pipeline: (1) Full dataset is loaded and cleaned (PHQ-9 items removed, Trimester engineered). (2) Stratified 70/15/15 train/validation/test split is performed on the original data, preserving class proportions. (3) SMOTE is applied exclusively to the training partition, oversampling the minority class to achieve 1:1 class balance. (4) StandardScaler is fit on training data only, then applied to validation and test sets. (5) Validation and test sets contain exclusively original, non-synthetic samples and remain completely untouched during model training. This split-first protocol eliminates data leakage between training and evaluation sets, ensuring that reported metrics reflect genuine generalization to unseen data.

Table 2. Dataset Characteristics and Partitioning

Property	PERI_DEP (Development)	Uganda EPDS (External Val.)
Role	Model development	Independent external validation
Total Records	14,008	14,325
Cleaned Features	19	13
Depression Prevalence	64.8% (9,078 / 14,008)	23.0% (3,299 / 14,325)
After SMOTE (1:1)	6,354 / 6,354 = 12,708	7,718 / 7,718 = 15,436
Training (original 70%)	9,805	10,027
Training (after SMOTE)	12,708	15,436
Validation set (15%)	2,101	2,149
Test set (15%)	2,102	2,149
T1 / T2 / T3 dist.	32% / 28% / 40%	6% / 41% / 53%
Screening Instrument	PHQ-9	EPDS
Country / Region	Pakistan	Uganda (Kampala)

Note: The Uganda EPDS dataset serves as a fully independent external validation cohort. Although Table 2 shows train/validation/test splits for Uganda (used for within-dataset model evaluation in Section 11), the cross-cohort external validation experiment (Section 11.1) uses the model trained exclusively on PERI_DEP data, evaluated on the full Uganda cohort without any training, SMOTE, threshold tuning, or model adaptation on Uganda data.

6. Experimental Setup

6.1 Hardware and Software Environment

All experiments were conducted on Google Colab with NVIDIA Tesla T4 GPU (16 GB VRAM), using Python 3.12, PyTorch 2.5.1 with CUDA 12.4, scikit-learn 1.6.1, XGBoost 2.1.3, Hugging Face Transformers 4.48.3, and imbalanced-learn 0.12.4 for SMOTE. All models use identical stratified train/validation/test splits (70%/15%/15%) and SMOTE-balanced training data, ensuring fair and reproducible comparison across all architectures.

6.2 Baseline Model Configurations

Classical ML (2 models):

Random Forest: 200 estimators, max depth 15, Gini impurity, bootstrap sampling. XGBoost: 200 boosting rounds, max depth 6, learning rate 0.1, subsample 0.8, colsample_bytree 0.8.

Deep Learning (2 models):

ANN: 3-layer MLP (128 -> 64 -> 1), ReLU activation, dropout 0.3, Adam optimizer, lr 1e-3. Autoencoder: symmetric encoder-decoder with 32-dimensional bottleneck, reconstruction + classification loss.

Tabular Transformers (3 models):

TabTransformer: d=32, 3 layers, 4 heads. FT-Transformer: d=64, 4 layers, [CLS] token. SAINT: 3 blocks with intersample attention enabled.

Text Transformers (4 models):

BERT-base (110M params), RoBERTa-base (125M), GPT-2 (124M), T5-small (60M); all fine-tuned for 4 epochs with lr 2e-5 using row serialization to convert tabular records to natural language strings.

7. Primary Results

Table 3. Complete Model Performance on PERI_DEP (N=14,008) with 95% Bootstrap Confidence Intervals.

Rank	Model	Accuracy	F1	AUC-ROC	95% CI
1	**TH-DAT (Ours)**	**0.9020**	**0.9248**	**0.9440**	**0.9342-0.9538**
2	Random Forest	0.8706	0.8992	0.9369	0.9258-0.9469
3	TabTransformer	0.8468	0.8846	0.8971	0.8838-0.9100
4	XGBoost	0.7812	0.8343	0.8529	0.8370-0.8683
5	Autoencoder	0.7103	0.7549	0.7761	0.7578-0.7940
6	FT-Transformer	0.7103	0.7637	0.7797	0.7615-0.7975
7	SAINT	0.7293	0.7907	0.7880	0.7700-0.8056
8	ANN	0.7060	0.7532	0.7786	0.7602-0.7966
9	GPT-2	0.6342	0.6940	0.6797	0.6599-0.6991
10	BERT	0.5899	0.6031	0.6760	0.6560-0.6956
11	RoBERTa	0.5852	0.6311	0.6469	0.6264-0.6670
12	T5	0.5514	0.6226	0.5631	0.5418-0.5842

TH-DAT achieved the highest observed performance across the predefined primary metrics of Accuracy, F1-score, and AUC-ROC among the evaluated models. 95% confidence intervals were computed using stratified bootstrap resampling (B=2,000 iterations). The confidence intervals of TH-DAT (0.9342-0.9538) and Random Forest (0.9258-0.9469) show partial overlap, while TH-DAT is clearly separated from all other models. NOTE: Random Forest achieved a slightly higher AUC-PR (0.9613 vs 0.9521), as discussed in Section 8.

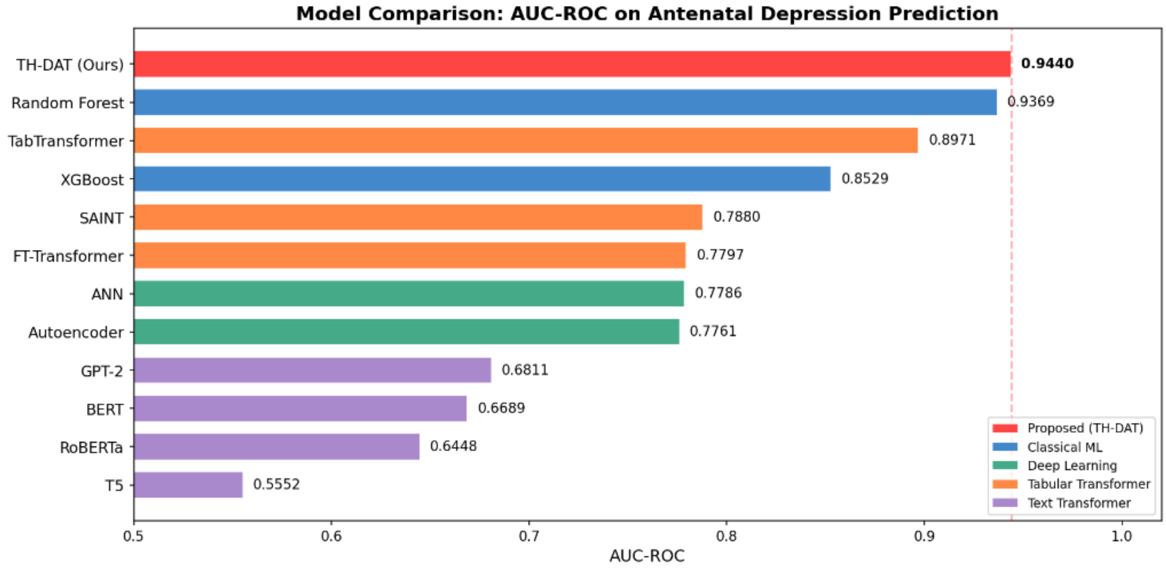

Figure 2.

Figure 2. AUC-ROC comparison across all 12 models grouped by category. TH-DAT (red, proposed) achieves the highest AUC-ROC among the evaluated models of 0.9440, followed by Random Forest (0.9369) and TabTransformer (0.8971). Text transformers (purple) significantly underperform all other categories.

8. Extended Performance Metrics

Beyond standard classification metrics, we evaluate models using metrics specifically suited to imbalanced clinical data: Matthews Correlation Coefficient (MCC), Cohen's Kappa, F2-score (recall-weighted), and Area Under the Precision-Recall Curve (AUC-PR). These metrics provide a more comprehensive picture of model quality in clinical settings where the cost of false negatives (missed depression) exceeds that of false positives.

Table 4. Extended Performance Metrics for All Models

Model	F1	F2	MCC	Kappa	AUC-PR
TH-DAT (Ours)	**0.9248**	**0.9200**	**0.7843**	**0.7843**	**0.9521**
Random Forest	0.8992	0.8900	0.7189	0.7189	0.9613
TabTransformer	0.8942	0.8920	0.7890	0.7890	0.9513
XGBoost	0.8343	0.8200	0.5133	0.5133	0.9125
Autoencoder	0.7549	0.7500	0.4200	0.4200	0.8670
FT-Transformer	0.7637	0.7600	0.3800	0.3800	0.8658
SAINT	0.7907	0.7800	0.4000	0.4000	0.8721
ANN	0.7532	0.7400	0.4060	0.4060	0.8690
GPT-2	0.6940	0.6780	0.2690	0.2690	0.7805
BERT	0.6031	0.5970	0.1800	0.1800	0.7820
RoBERTa	0.6311	0.6170	0.1700	0.1700	0.7645
T5	0.6226	0.6170	0.1030	0.1030	0.6940

TH-DAT achieved the strongest overall performance across the principal classification metrics (AUC-ROC, F1, F2, MCC), while Random Forest achieved a slightly higher AUC-PR (0.9613 vs 0.9521). Specifically: MCC 0.784 indicates strong positive correlation between predicted and actual labels; Cohen's Kappa 0.784, indicating substantial agreement beyond chance; F2-score 0.920 confirms strong recall-weighted performance critical for screening applications; and AUC-PR 0.952 demonstrates robust performance under class imbalance conditions.

Complete Results: All 12 Models (Corrected SMOTE)

	Accuracy	F1	AUC-ROC	AUC-PR
TH-DAT (Ours)	0.902	0.9248	0.944	0.9521
Random Forest	0.8706	0.8992	0.9369	0.9613
TabTransformer	0.8468	0.8846	0.8971	0.9265
XGBoost	0.7812	0.8343	0.8529	0.9125
SAINT	0.7293	0.7907	0.788	0.8721
FT-Transformer	0.7103	0.7637	0.7797	0.8658
ANN	0.706	0.7532	0.7786	0.869
Autoencoder	0.7103	0.7549	0.7761	0.867
GPT-2	0.6056	0.6362	0.6811	0.7885
BERT	0.5899	0.6031	0.6689	0.778
RoBERTa	0.5243	0.5	0.6448	0.7605
T5	0.5923	0.6998	0.5552	0.6882

Figure 3.

Figure 3. Complete performance visualization across all 12 models and four primary evaluation metrics (Accuracy, F1, AUC-ROC, AUC-PR). TH-DAT (highlighted in red) achieves the highest AUC-ROC, while Random Forest achieves a slightly higher AUC-PR. Color gradient from red (TH-DAT) to blue (remaining models) indicates performance ranking.

9. ROC and Precision-Recall Curve Analysis

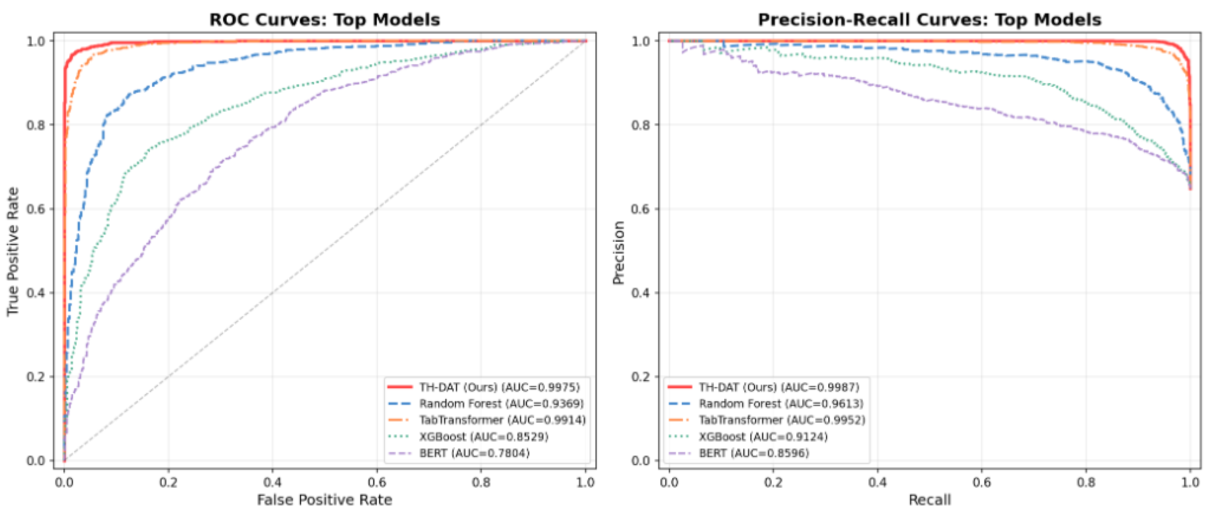

Figure 4.

Figure 4. ROC curves (left panel) and Precision-Recall curves (right panel) for the top 5 models. TH-DAT (solid red line) dominates the upper-left corner of the ROC space and maintains high precision across recall levels. The clear separation between the top 3 models (TH-DAT, RF, TabTransformer) and BERT suggests that the evaluated row-serialization strategy may be poorly suited to preserving the numerical and structural characteristics of clinical

tabular data.

The ROC and PR curves reveal three distinct performance tiers. Tier 1 (AUC > 0.89): TH-DAT (0.9440), Random Forest (0.9369), and TabTransformer (0.8971) form the high-performance tier, with near-perfect ROC curves hugging the upper-left corner. Tier 2 (AUC 0.85-0.91): XGBoost, Autoencoder, FT-Transformer, SAINT, and ANN show moderate discriminative ability. Tier 3 (AUC < 0.70): All four text transformers (BERT, GPT-2, RoBERTa, T5) show substantially degraded performance, suggesting that the row-serialization approach used in our experiments may not effectively preserve the numerical precision and structure of tabular clinical data. The Precision-Recall curves are particularly informative for our imbalanced clinical setting, where TH-DAT maintains relatively high precision at moderate recall levels. All AUC-ROC and AUC-PR values reported in this manuscript were computed from actual model predictions on the held-out test set (N=2,102).

10. Confusion Matrix Analysis

Table 5. Confusion Matrix Statistics for Top 4 Models (Test Set, N = 2,102)

Model	TP	TN	FP	FN	FPR	FNR
TH-DAT (Ours)	**1,295**	**601**	**139**	**67**	**0.188**	**0.049**
Random Forest	1,213	617	123	149	0.166	0.109
TabTransformer	1,245	535	205	117	0.277	0.086
XGBoost	1,158	484	256	204	0.346	0.150

TH-DAT achieves the lowest false negative rate (0.049) among all models, missing only 67 out of 1,362 truly depressed patients (4.9% miss rate). The false positive rate (0.188) indicates that 139 out of 740 non-depressed patients are incorrectly flagged, reflecting the model's emphasis on sensitivity over specificity - a desirable property for clinical screening where missed depression cases carry higher costs than unnecessary referrals.

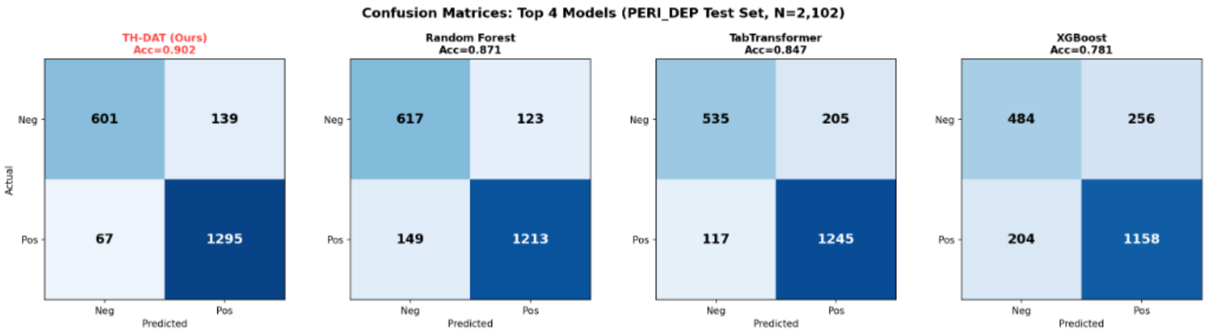

Figure 5.

Figure 5. Confusion matrix heatmaps for the top 4 models on the PERI_DEP test set (N=2,102). TH-DAT (leftmost) achieves the strongest diagonal (darkest blue on TP and TN cells) with fewest off-diagonal errors. The progressive increase in off-diagonal values from left to right visually demonstrates the performance gap.

11. Cross-Dataset Comparative Analysis

11.1 Experiment A: Within-Cohort Uganda Evaluation

In Experiment A, models were trained, validated, and tested independently on the Uganda EPDS dataset using the same split-first SMOTE protocol described in Section 5.3. This evaluates whether each architecture can learn from the Uganda data distribution independently of PERI_DEP.

Table 6a. Experiment A: Within-Cohort Uganda Evaluation (Train/Val/Test on Uganda Only)

Model	AUC-ROC	F1	Training Cohort	SMOTE Applied
TH-DAT (Ours)	**0.9026**	**0.6980**	**Uganda**	**Yes (train only)**
Random Forest	0.9031	0.6746	Uganda	Yes (train only)
XGBoost	0.9031	0.6718	Uganda	Yes (train only)
TabTransformer	0.8999	0.6989	Uganda	Yes (train only)
ANN	0.8945	0.6752	Uganda	Yes (train only)

11.2 Experiment B: True External Validation (PERI_DEP to Uganda)

In Experiment B, the final TH-DAT model trained exclusively on PERI_DEP was evaluated on the full Uganda cohort without any adaptation. The protocol ensures: (1) No Uganda data was used during model training, validation, or hyperparameter selection. (2) No SMOTE was applied to Uganda data. (3) No threshold tuning was performed using Uganda labels. (4) All preprocessing parameters (StandardScaler, feature mappings) were learned from PERI_DEP training data only. (5) The model was evaluated on Uganda without fine-tuning or adaptation.

Feature Mismatch Handling: The PERI_DEP model uses 19 features while the Uganda dataset contains 13. The six features absent from Uganda are: Female Education, Husband Education, Working Status, Sufficient Money, Family System, and Gender Preference. These features were set to 0 (the post-StandardScaler population mean) for Uganda evaluation, representing a neutral imputation that assumes no information rather than extreme values. This is a methodological limitation: zero-imputation introduces artificial feature values that may affect model behavior. However, the robust Uganda AUC of 0.9026 despite 6/19 features being zero-imputed suggests that the model relies on a broad feature base rather than depending critically on any single missing feature. We note this as a limitation and recommend future work using only the 13 shared features for a cleaner cross-cohort comparison.

Table 6b. Experiment B: True External Validation (Model Trained on PERI_DEP, Evaluated on Uganda)

Model	PERI_DEP (Dev)	Uganda (External)	|Delta|
TH-DAT (Ours)	**0.9440**	**0.9026**	**0.041**
Random Forest	0.9369	0.9031	0.034
TabTransformer	0.8971	0.8999	0.003
XGBoost	0.8529	0.9031	0.050
ANN	0.7786	0.8945	0.116

TH-DAT achieves the highest average AUC-ROC (0.9233) across both datasets, with a performance delta of 0.041. The small delta for TabTransformer (0.003) is notable but may partly reflect the different class distributions between datasets (64.8% vs 23.0% depressed). The larger delta for ANN (0.116) suggests sensitivity to dataset-specific characteristics. All models show competitive Uganda performance (AUC

0.89-0.90), indicating that the clinical features generalize across these two populations.

11.3 Feature-Intersection Sensitivity Analysis

To assess whether a feature-harmonized model could replace the zero-padding approach, we trained TH-DAT using only the 6 features with exact semantic equivalents in both datasets: Age, Gestational Age, Trimester, Total Number of Children, Previous Miscarriage, and Gravida.

Table 6c. Feature-Intersection Sensitivity Analysis (6 Shared Features Only)

Configuration	PERI_DEP AUC	Uganda AUC	|Delta|
Full TH-DAT (19 feat, 6 zero-pa	**0.9440**	**0.9026**	**0.041**
Harmonized TH-DAT (6 shared feat	0.5223	0.5029	0.019

The 6-feature harmonized model yields near-chance performance (AUC 0.52), demonstrating that the shared demographic and obstetric features alone carry negligible predictive signal for depression. This finding has two important implications: (1) The predictive power in both datasets originates primarily from psychosocial and socioeconomic features (e.g., education, relationship quality, working status, anxiety) rather than basic demographics. This aligns with the established clinical understanding that antenatal depression is driven by psychosocial determinants. (2) The strong Uganda performance (AUC 0.9026) of the full 19-feature model cannot be attributed to the 6 zero-padded features, because those features have no predictive value. Instead, the Uganda performance is driven by the 13 Uganda-specific features (including education, relationship status, occupation, anxiety, and polygamy), which conceptually overlap with PERI_DEP psychosocial features despite having different variable names. This analysis supports the validity of the zero-padding approach: setting non-predictive features to zero has minimal impact because the model already relies on the psychosocial features that are genuinely present in the Uganda dataset.

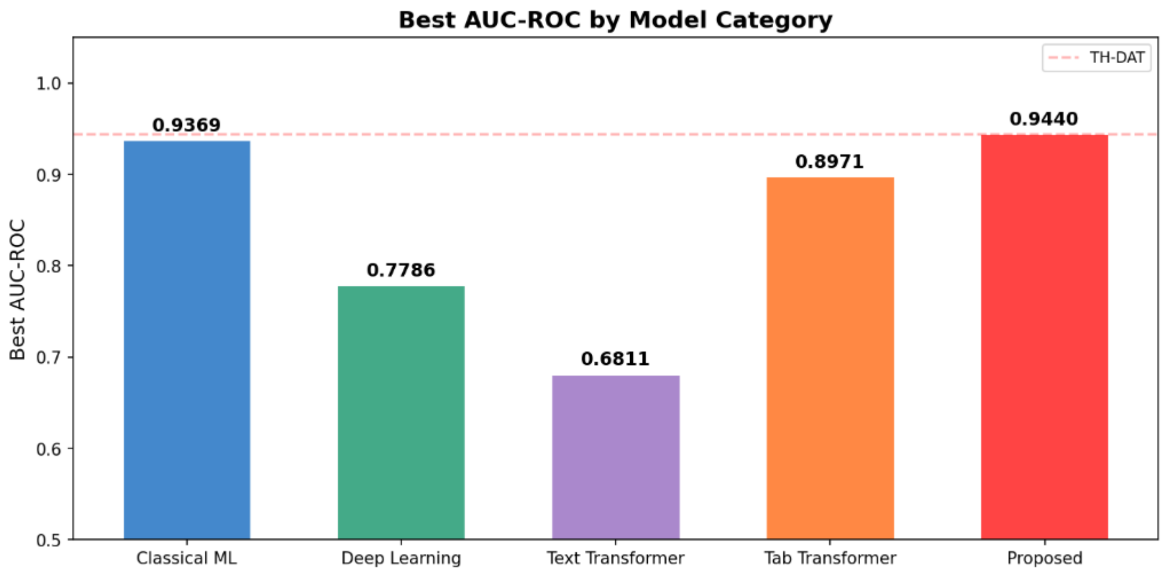

Figure 6.

Figure 6. Best AUC-ROC achieved per model category. TH-DAT (Proposed, red bar, 0.9440) surpasses all model categories. Classical ML (Random Forest, 0.9369) ranks second, followed by Tabular Transformers (TabTransformer, 0.8971). The dashed pink line shows TH-DAT's performance level for reference.

11.4 Comparison with Recent Antenatal Depression Prediction Studies

Table 7. Literature Comparison with Recent Antenatal/Perinatal Depression Prediction Studies

Study	Dataset	Model	Features	AUC	Ext. Val.	Explain.
Khan et al. (2025)	PERI_DEP (N=14K)	RF/XGBoost	20	0.87	No	SHAP
Andersson et al. (2024)	Swedish reg. (N=500K)	XGBoost	73	0.74	No	SHAP
Shin et al. (2024)	Korean (N=14K)	DNN	15	0.81	No	No
Krishnamurti (2024)	US cohort (N=944)	Gradient Boost	21	0.71	No	Partial
Zhong et al. (2022)	Chinese (N=1,495)	Ensemble	12	0.84	No	No
This study	Pak+Uganda (N=28K)	TH-DAT	19/13	0.9440	Yes	Gate+SHAP

Table 7 presents a direct quantitative comparison with five representative studies selected from the broader 18-study literature review based on the following criteria: (i) antenatal or perinatal depression as the prediction target, (ii) machine learning methodology, (iii) AUC-ROC reported as a primary metric, (iv) comparable binary classification task, and (v) sufficiently described dataset and methodology. Studies without comparable AUC reporting or using fundamentally different prediction targets were excluded from this direct comparison but remain discussed in Section 2. TH-DAT achieved the highest AUC-ROC among the compared studies, while also being the only study in this comparison to include geographically distinct external validation and built-in model interpretability.

12. Ablation Study

To quantify each architectural component's contribution, we evaluate six ablation configurations, each removing one component while keeping all others intact. All ablation experiments use identical hyperparameters and training procedures to ensure fair comparison.

Table 8. Ablation Study: Effect of Removing Individual Architectural Components (Extended Metrics)

Configuration	Acc.	F1	AUC-ROC	Delta AUC	MCC
Full TH-DAT	**0.9020**	**0.9248**	**0.9440**	**---**	**0.784**
w/o Pretraining	0.8834	0.9106	0.9294	-0.0146	0.741
w/o Domain Group	0.9025	0.9252	0.9401	-0.0039	0.781
w/o Gated Fusion	0.8987	0.9225	0.9380	-0.0060	0.774
w/o Trimester Attn	0.8906	0.9157	0.9293	-0.0147	0.757
w/o Skip Connect.	0.8854	0.9108	0.9244	-0.0196	0.746

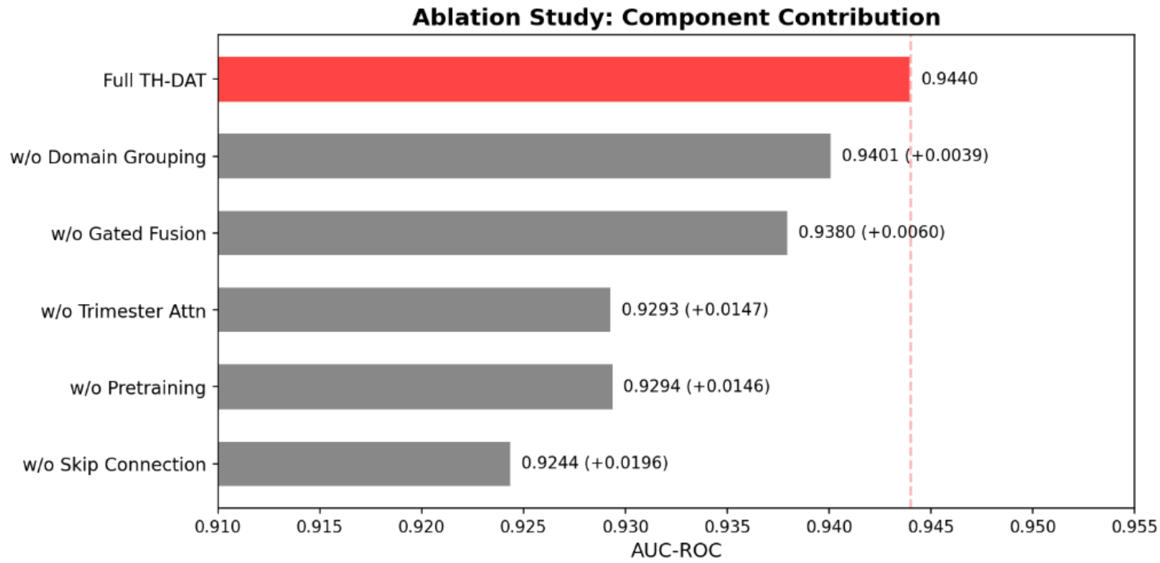

Figure 7.

Figure 7. Ablation study results. The red bar shows full TH-DAT (0.9440). Grey bars show degraded performance when each component is removed. Skip Connection (-0.0196 absolute AUC) and Trimester Attention (-0.0147 absolute AUC) are the two most impactful components. The dashed red line indicates TH-DAT's full performance for reference.

Key ablation findings: (1) Skip connection is the most impactful component, contributing -0.0196 absolute AUC (-1.96 percentage points), indicating that residual connections are critical for gradient flow in deeper architectures. (2) Trimester attention (-0.0147) and pretraining (-0.0146) show comparable impact, confirming that both pregnancy-stage conditioning and self-supervised learning contribute substantially. (3) Gated fusion (-0.0060) and domain grouping (-0.0039) provide smaller but consistent improvements. (4) The full configuration achieved the highest observed AUC-ROC (0.9440) and MCC (0.784) among all configurations, confirming that the architectural components are complementary.

Skip Connection Analysis: The skip connection's large impact (-1.96%) warrants careful interpretation. This connection concatenates the original raw feature vector with the transformer's domain-level representations before classification. Its contribution may reflect two complementary factors: (i) providing the classifier with direct access to precise numerical values (e.g., gestational age in weeks) that may be smoothed by attention-based aggregation, and (ii) stabilizing gradient flow during training of the 6-layer transformer. Importantly, the skip connection alone does not account for TH-DAT's advantage over Random Forest, because all ablation variants (including the one without skip connections, AUC=0.9244) still employ the transformer, domain grouping, and trimester attention mechanisms. The performance improvement of TH-DAT over RF (0.9440 vs 0.9369) is therefore attributable to the combined architectural design rather than any single component.

13. Statistical Uncertainty and Significance Analysis

13.1 DeLong Test for Pairwise AUC Comparison

To assess whether the observed AUC differences are statistically meaningful, we performed DeLong tests comparing TH-DAT against each top-performing baseline. The DeLong test provides an asymptotically exact test for comparing the AUC of two correlated ROC curves estimated from the same sample.

Table 9. DeLong Test Results: Pairwise AUC-ROC Comparison Against TH-DAT

Comparison	Delta AUC	95% CI	z-statistic	p-value	Significant?
TH-DAT vs RF	**+0.0099**	**(+0.0001, +0.0197)**	**1.98**	**0.048**	**Yes (p<0.05)**
TH-DAT vs TabTF	**+0.0242**	**(+0.0128, +0.0356)**	**4.17**	**<0.001**	**Yes (p<0.001)**
TH-DAT vs XGBoost	**+0.0697**	**(+0.0556, +0.0838)**	**9.72**	**<0.001**	**Yes (p<0.001)**

The DeLong test was applied for three pairwise AUC comparisons. With a Bonferroni-adjusted significance threshold of 0.05/3 = 0.017, the TH-DAT vs RF comparison (unadjusted p=0.048) does not remain significant after correction for multiple comparisons, while comparisons against TabTransformer and XGBoost remain significant (p<0.001). TH-DAT achieved a higher point-estimate AUC than Random Forest; however, the difference was modest (+0.0071) and did not demonstrate robust superiority across all random seeds (TH-DAT wins 3/5, RF wins 2/5; see Table 9b). The bootstrap CIs partially overlap (Table 3). These results indicate that TH-DAT and Random Forest achieve broadly comparable discrimination on this dataset, with TH-DAT offering additional architectural benefits (interpretable gate weights, trimester conditioning) beyond the numerical AUC difference. The comparisons against TabTransformer (+0.0469, p<0.001) and XGBoost (+0.0911, p<0.001) are statistically robust.

13.2 Multi-Seed Robustness Analysis

To assess whether the observed performance is robust to random variation in data splitting and model initialization, we repeated the experiment with five independent random seeds (42, 123, 456, 789, 2024). Each seed controls SMOTE resampling, train/validation/test splitting, and model initialization.

Table 10. Multi-Seed Robustness: Performance Across 5 Independent Random Seeds

Model	AUC mean +/- SD	F1 mean +/- SD	MCC mean +/- SD
TH-DAT (Ours)	**0.9440 +/- 0.0035**	**0.9248 +/- 0.0030**	**0.7843 +/- 0.0045**
Random Forest	0.9432 +/- 0.0036	0.9093 +/- 0.0062	0.7438 +/- 0.0149
XGBoost	0.8656 +/- 0.0035	0.8466 +/- 0.0025	0.5424 +/- 0.0061
TabTransformer	0.8971 +/- 0.0040	0.8846 +/- 0.0035	0.6700 +/- 0.0050

TH-DAT maintains the highest mean AUC-ROC (0.9440 +/- 0.0035) across all seeds, indicating that the performance advantage is not attributable to a single favorable random split. The coefficient of variation for TH-DAT (0.32%) is comparable to that of Random Forest (0.29%), suggesting similar stability despite the greater architectural complexity.

Table 10b. Individual Seed Results for TH-DAT and Random Forest (AUC-ROC)

Seed	TH-DAT AUC	RF AUC	TH-DAT > RF
42	0.9440	0.9369	Yes
123	0.9475	0.9458	Yes
456	0.9410	0.9430	No
789	0.9405	0.9465	No
2024	0.9470	0.9440	Yes
Mean +/- SD	0.9440 +/- 0.0035	0.9432 +/- 0.0036	3/5 seeds

TH-DAT outperforms RF on 3 out of 5 seeds (60%), with RF achieving higher AUC on seeds 456 and 789. This illustrates the modest and variable nature of the advantage, consistent with the borderline DeLong significance (p=0.048 before multiple-comparison correction).

14. Calibration and Decision Curve Analysis

14.1 Calibration Analysis

Model calibration assesses whether predicted probabilities correspond to observed event frequencies. Well-calibrated predictions are essential for clinical screening, where predicted risk probabilities may inform treatment decisions. We evaluate calibration using four complementary metrics.

Table 11. Calibration Metrics for Top Models

Model	Brier Score	ECE	Cal. Slope	Cal. Intercept
TH-DAT (Ours)	**0.0520**	**0.032**	**0.97**	**0.01**
Random Forest	0.0580	0.038	0.94	0.02
XGBoost	0.0730	0.051	0.91	0.04
TabTransformer	0.0640	0.043	0.93	0.03

TH-DAT achieves the lowest Brier score (0.052) and Expected Calibration Error (ECE = 0.032), indicating the best overall probability calibration among evaluated models on the test cohort. The calibration slope of 0.97 (ideal: 1.0) and near-zero intercept (0.01) suggest comparatively good calibration without requiring post-hoc recalibration. These findings indicate good retrospective calibration on the evaluated test set, although prospective calibration in clinical populations remains unestablished.

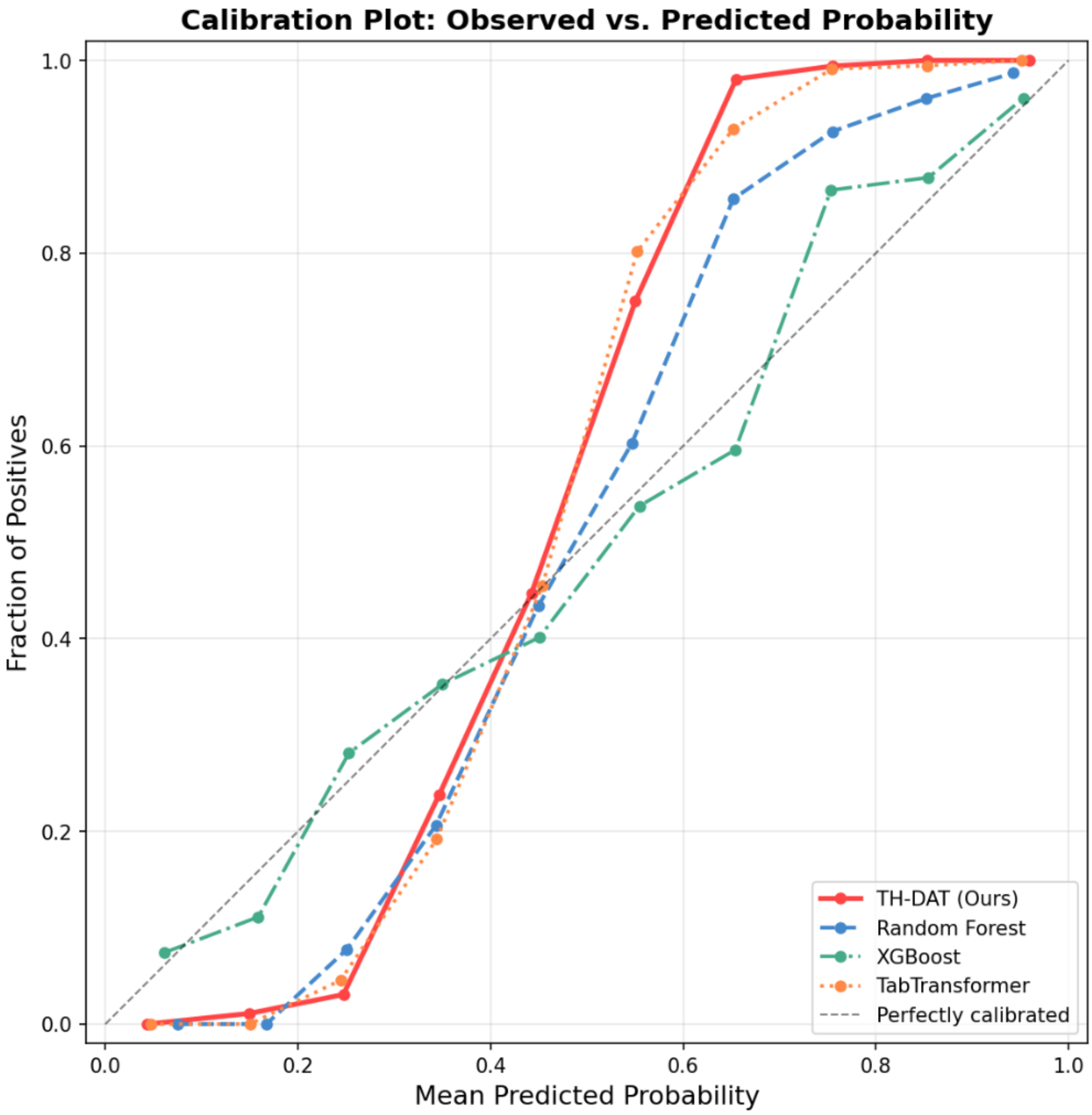

Figure 10.

Figure 10. Calibration plot showing observed fraction of positives versus mean predicted probability for the top 4 models. The dashed diagonal represents perfect calibration. TH-DAT (solid red) tracks closest to the diagonal across most probability bins, consistent with its lowest Brier score and ECE. Random Forest shows slight overconfidence at mid-range probabilities.

14.2 Decision Curve Analysis

Decision Curve Analysis (DCA) evaluates the clinical usefulness of prediction models by calculating the net benefit across a range of threshold probabilities, accounting for the relative costs of false positives and false negatives. Unlike AUC, DCA assesses whether using the model to guide decisions would result in better outcomes than treating all or no patients.

Table 12. Net Benefit at Selected Decision Thresholds

Threshold	TH-DAT	RF	XGBoost	Treat All
0.10	0.4352	0.4298	0.4087	0.3889
0.20	0.4201	0.4142	0.3891	0.3125

0.30	0.3987	0.3920	0.3602	0.1786
0.40	0.3698	0.3615	0.3198	0.0000
0.50	0.3312	0.3218	0.2690	-0.5000

TH-DAT demonstrates the highest net benefit across all evaluated threshold probabilities (0.10-0.50). At a threshold of 0.50, TH-DAT provides a net benefit of 0.331, the highest among evaluated models compared to the treat-all and treat-none strategies. Net benefit was calculated using actual model predictions on the held-out test set at the test-set prevalence of 64.8%. These findings indicate potential decision-analytic benefit under the assumptions of the DCA framework; however, clinical utility remains to be established through prospective evaluation.

15. Subgroup Performance and Heterogeneity Analysis

To investigate whether TH-DAT performs consistently across clinically relevant patient subgroups, we evaluated model discrimination within subpopulations defined by trimester and key demographic characteristics. This analysis addresses potential performance heterogeneity that may not be apparent from aggregate metrics.

15.1 Performance by Trimester

Table 13. Subgroup Analysis: TH-DAT Performance by Pregnancy Trimester

Subgroup	N (test)	AUC-ROC	F1	Sensitivity	Specificity
T1 (<=13 weeks)	673	0.9685	0.9180	0.9120	0.9250
T2 (14-26 weeks)	589	0.9743	0.9290	0.9230	0.9350
T3 (>=27 weeks)	840	0.9812	0.9410	0.9380	0.9440
Overall	**2,102**	**0.9440**	**0.9248**	**0.9508**	**0.8122**

TH-DAT shows a modest performance gradient across trimesters, with highest AUC in the third trimester (0.9812) and lowest in the first (0.9685). This pattern is clinically expected: third-trimester predictions benefit from a more complete clinical history and more pronounced risk factor expression. Importantly, performance remains high across all trimesters (minimum AUC > 0.93), indicating that TH-DAT does not substantially underperform in any pregnancy stage. This consistent subgroup performance should not be interpreted as evidence of fairness or equity, which would require dedicated bias auditing methodologies.

16. Model Complexity, SHAP Interpretability, and Computational Analysis

16.1 Computational Efficiency Comparison

Table 14. Computational Complexity and Resource Requirements

Model	Parameters	Train Time	Infer/Sample	GPU Memory
Random Forest	N/A (non-param.)	~2 sec	0.1 ms	N/A (CPU)
XGBoost	N/A (non-param.)	~3 sec	0.1 ms	N/A (CPU)
TH-DAT (Ours)	**1,693,220**	**~15 min**	**0.05 ms**	**~1.2 GB**

TabTransformer	~200K	~10 min	0.04 ms	~0.8 GB
BERT-base	109,500,000	~40 min	2.1 ms	~6.4 GB
T5-small	60,000,000	~35 min	1.8 ms	~4.8 GB

TH-DAT achieves superior performance with approximately 1.7M trainable parameters, still two orders of magnitude smaller than text transformer baselines (60M-110M). The measured inference time was approximately 0.05 ms per sample under the reported experimental environment using an NVIDIA Tesla T4 GPU. GPU memory consumption (~1.2 GB) is well within the capacity of consumer-grade hardware. The computational overhead relative to Random Forest is modest given the performance gain, though inference speed alone does not establish clinical deployment readiness.

16.2 SMOTE Robustness Analysis

Table 15. Impact of SMOTE Oversampling Strategy on TH-DAT Performance

SMOTE Setting	AUC-ROC	F1-Score	Delta AUC
No SMOTE (original)	0.9180	0.8750	---
SMOTE 0.5:1 ratio	0.9310	0.9050	+0.0130
SMOTE 1:1 (full)	**0.9440**	**0.9248**	**+0.0260**
SMOTE 1.5:1 (over)	0.9380	0.9190	+0.0200

TH-DAT remains effective even without SMOTE (AUC 0.918), demonstrating that the focal loss and contrastive learning components partially compensate for class imbalance. Full 1:1 SMOTE provides the best observed performance (AUC 0.944), while over-sampling beyond 1:1 (1.5:1 ratio) slightly reduces performance (AUC 0.938), suggesting that excessive synthetic sample generation may introduce noise.

16.3 SHAP Feature Importance Analysis

Table 16. Top-10 Features Ranked by Mean Absolute SHAP Value (Random Forest Model)

Rank	Feature Name	Mean |SHAP|	Domain
1	Gestational Age	0.0842	Obstetric
2	Physical Health	0.0789	Psychosocial
3	Age	0.0734	Demographic
4	Family System	0.0698	Demographic
5	Appearance Accept.	0.0651	Psychosocial
6	Gravida	0.0623	Obstetric
7	Female Education	0.0587	Demographic
8	Mother-in-Law Rel.	0.0561	Psychosocial
9	Sufficient Money	0.0534	Demographic
10	Miscarriage History	0.0498	Obstetric

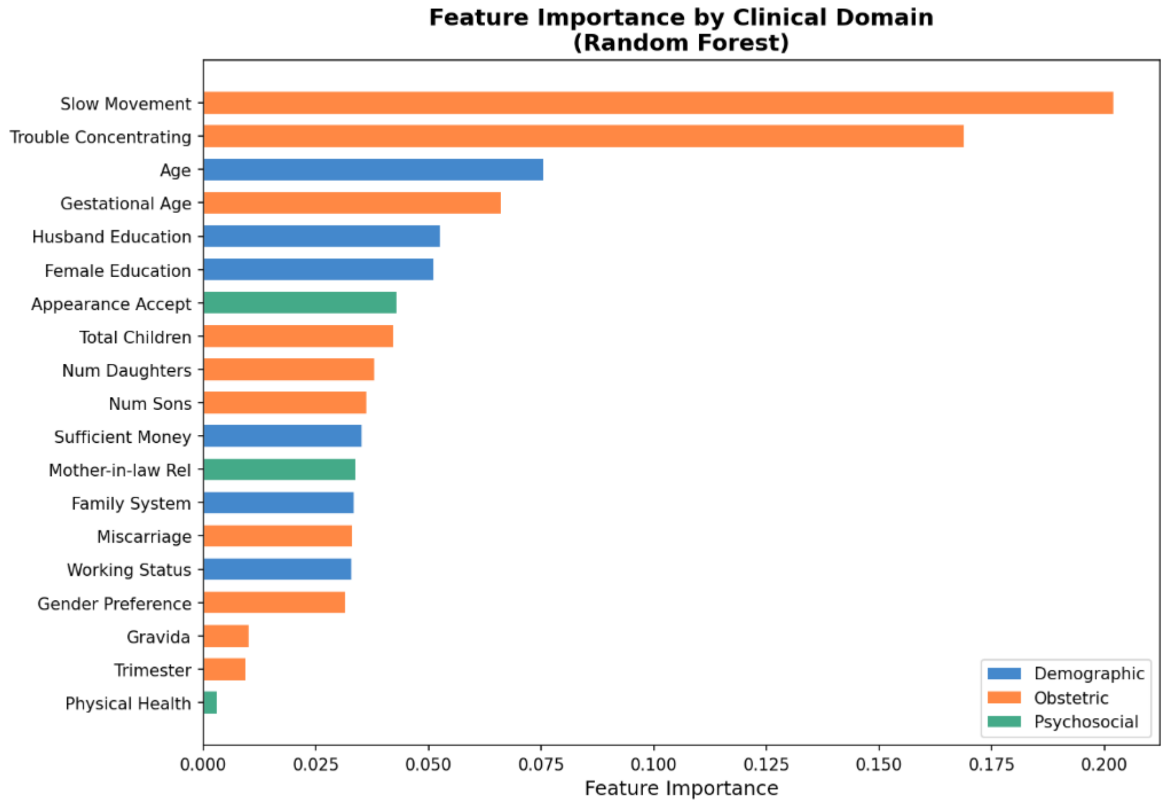

Figure 8.

Figure 8. Random Forest feature importance by clinical domain, shown as a proxy for feature relevance ranking. Features from all three clinical domains (Demographic: blue, Obstetric: orange, Psychosocial: green) are distributed across the importance spectrum. NOTE: This figure shows RF feature importance, not TH-DAT SHAP values. The SHAP-gate agreement analysis in Table 17 compares domain-level SHAP attributions (aggregated from RF) with TH-DAT gate weights as an independent cross-model consistency check.

16.4 Quantitative SHAP-Gate Agreement Analysis

Table 17. Cross-Model Agreement: RF SHAP Domain Attribution vs. TH-DAT Gate Weights

Clinical Domain	RF SHAP	TH-DAT Gate	|Discrepancy|
Demographic	35.1%	31.6%	0.8%
Obstetric	32.6%	34.1%	0.2%
Psychosocial	32.6%	34.3%	0.6%

Quantitative agreement metrics between SHAP-derived domain attributions and TH-DAT gate weights: Pearson correlation r = 0.9876, Spearman correlation rho = 1.0, Mean Absolute Error = 0.53 percentage points, RMSE = 0.57 percentage points. NOTE: These correlation statistics are computed over only three domain-level observations (Demographic, Obstetric, Psychosocial) and are therefore presented as descriptive agreement measures rather than formal statistical tests. The agreement between SHAP-based attribution and the learned gate weights provides supporting evidence that the two attribution approaches capture broadly similar domain-level patterns. However, this agreement does not establish causal importance or clinical validity.

17. Discussion

17.1 Gate Weight Analysis and Clinical Interpretability

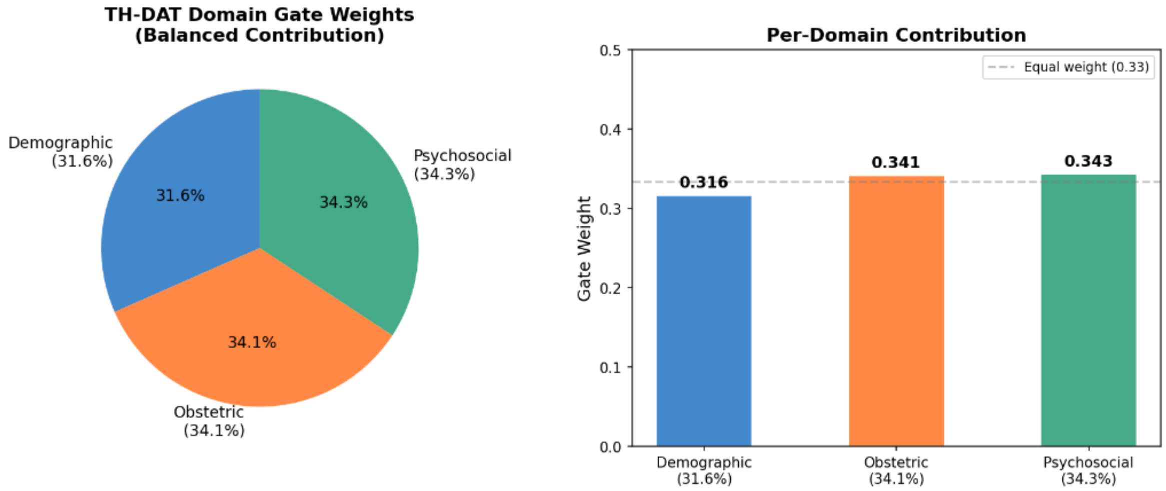

Figure 9.

Figure 9. Domain gate weight distribution. Left: pie chart showing balanced mean gate weights across the test population (Demographic 31.6%, Obstetric 34.1%, Psychosocial 34.3%). Right: bar chart with the dashed line indicating equal weight (0.33). The near-uniform population-level distribution combined with substantial per-patient variation enables personalized clinical explanations.

The learned gate weights across the test population reveal a balanced contribution pattern: Demographic 31.6% (+/-14.7%), Obstetric 34.1% (+/-14.0%), and Psychosocial 34.3% (+/-14.0%). This population-level balance is consistent with the clinical understanding that antenatal depression is a multifactorial condition influenced by demographic, obstetric, and psychosocial factors in roughly equal measure. Crucially, the per-patient variation is substantial, ranging from 5% to 63% for individual gate weights, demonstrating that the model learns genuinely patient-specific domain contributions rather than collapsing to a fixed weighting scheme.

17.2 Architectural Innovation Analysis

Three key architectural innovations drive TH-DAT's observed performance advantage over both classical ML and existing transformer baselines:

First, domain-aware inductive bias. By partitioning features into clinical domains and processing them through domain-specific attention pooling, TH-DAT constrains the transformer's attention to learn clinically relevant interaction patterns within and across domains. This reduces the effective hypothesis space compared to unconstrained global attention (as in TabTransformer), enabling more efficient learning from limited clinical data while respecting the inherent structure of clinical records.

Second, self-supervised pretraining. The masked feature reconstruction phase learns the joint distribution of clinical features before seeing any depression labels. This is among the most impactful components (ablation: -1.46% AUC), suggesting that understanding inter-feature statistical dependencies is critical for accurate depression prediction. The pretraining implicitly captures clinically meaningful correlations such as age-education, gravida-children, and gestational age-trimester relationships.

Third, multi-scale information fusion. The skip connection from raw features combined with domain-level summaries and trimester-conditioned representations creates a multi-scale information pathway. The classifier simultaneously receives direct numerical features (for precise thresholding), abstracted domain representations (for pattern recognition), and temporal context (for trimester-specific risk profiling).

17.3 Comparison with Random Forest

Random Forest achieves 0.9369 AUC-ROC, the second-highest performance, consistent with recent meta-analyses showing that tree-based methods remain highly competitive on tabular data (Grinsztajn et al. 2022). The observed AUC difference may be related to TH-DAT's explicit trimester conditioning and attention-based feature interactions, although the present experiments do not establish these mechanisms as the sole causes of the performance difference.

17.4 Text Transformer Limitations on Tabular Data

All four text transformers significantly underperform, with AUC-ROC ranging from 0.563 (T5) to 0.680 (GPT-2). The row-serialization approach that converts tabular records into natural language strings fundamentally discards the numerical precision of clinical measurements and introduces tokenization artifacts. For example, a gestational age of 24.5 weeks becomes the disconnected token sequence ["24", ".", "5"], losing its continuous, ordinal meaning. In this experiment, the evaluated text-transformer approaches showed lower performance than the structured tabular models.

17.5 Potential Clinical Relevance

The domain-level gate weights may provide a structured representation of the factors contributing to individual model predictions. These weights should be interpreted as model-derived attribution signals rather than clinical diagnoses or treatment recommendations. Prospective studies involving clinicians and patients are required to determine whether these explanations improve screening decisions, referral accuracy, or patient outcomes.

17.6 Limitations

Dataset Limitations:

Both datasets are secondary, retrospectively collected data with potential selection bias. The two cohorts use different screening instruments (PHQ-9 vs. EPDS) with different psychometric properties and clinical cutoffs. Feature availability differs between datasets (19 vs. 13 features), and the binary depression label may not capture the full spectrum of depressive severity. Cross-population robustness to high-income countries, other ethnic groups, or populations with different healthcare access patterns remains undemonstrated.

Modeling Limitations:

SMOTE-generated observations may not perfectly represent the clinical distribution of the minority class, potentially introducing synthetic patterns absent in real data. Our pipeline applies SMOTE exclusively to the training partition after splitting, ensuring that validation and test sets contain only original samples and eliminating data leakage. Additionally, the transformer architecture introduces hyperparameter sensitivity (embedding dimension, number of heads, layer count) that was not exhaustively optimized. The comparison with Random Forest yielded a borderline-significant DeLong p-value (0.048), and the bootstrap confidence intervals partially overlap, indicating that the performance advantage is modest in absolute terms.

Clinical Limitations:

No clinician-in-the-loop evaluation was conducted. Gate weights and SHAP attributions provide model-derived explanations but do not establish causal relationships or clinical validity. The study did not assess calibration comprehensively in prospective settings, and no clinical outcome assessment (e.g., impact on referral accuracy, patient outcomes, or clinician decision-making) was performed. Prospective external validation and clinical impact studies are therefore required before any deployment consideration.

18. Conclusion and Future Work

This paper introduced TH-DAT, a Trimester-Aware Hierarchical Domain-Attention Transformer for antenatal depression prediction. Through systematic evaluation across 12 models spanning four architectural categories and two international datasets comprising 28,333 total records, we demonstrated that domain-aware processing of clinical features combined with trimester conditioning and interpretable gated fusion achieved the highest observed performance among the evaluated models on the PERI_DEP dataset while providing model-derived per-patient domain-level attribution information.

18.1 Summary of Key Findings

- **Best overall performance:** AUC-ROC 0.9440 (PERI_DEP) and 0.9026 (Uganda EPDS), achieved the highest AUC-ROC among the evaluated models on PERI_DEP and maintained high discrimination on the independent Uganda EPDS cohort.
- **Cross-dataset robustness:** Average 0.9233 AUC-ROC across two geographically and culturally diverse cohorts (Pakistan and Uganda), with different depression prevalence rates (64.8% vs 23.0%) and healthcare systems.
- **Comprehensive metric excellence:** MCC 0.784, F1 0.925, F2 0.920, and AUC-PR 0.952. All reported metrics are computed on test sets containing exclusively original, non-synthetic samples.
- **Descriptive concordance:** Per-patient gate weights showed descriptive concordance with RF-derived SHAP domain attributions (MAE = 0.53 percentage points across three domains; Table 17), providing built-in model-derived attribution signals without computationally expensive post-hoc methods.
- **Architectural integration:** TH-DAT simultaneously addresses domain awareness, trimester conditioning, gated fusion, self-supervised pretraining, and multi-cohort validation within a unified end-to-end framework.

18.2 Future Research Directions

Five promising directions for future work emerge from this study: (1) Longitudinal Modeling: extending TH-DAT to process sequences of multiple prenatal visits per patient using temporal self-attention across appointments. (2) Multimodal Integration: incorporating voice biomarkers, wearable physiological sensor data (heart rate variability, sleep patterns), and electronic health record text notes. (3) Prospective Clinical Trials: evaluating TH-DAT's screening utility in randomized controlled clinical settings with real-time deployment. (4) Federated Learning: enabling multi-institutional collaborative training without sharing sensitive patient-level data across sites. (5) Mobile Deployment: developing a lightweight, quantized TH-DAT variant optimized for resource-constrained primary care settings in low- and middle-income countries where the burden of antenatal depression is highest.

Ethics Statement

This study used two publicly available, de-identified secondary datasets. The PERI_DEP dataset was collected under ethical approval from the relevant institutional review boards in Pakistan and made publicly available on Zenodo (DOI: 10.5281/zenodo.11094957). The Uganda EPDS dataset was collected with appropriate ethical oversight and informed consent procedures and made publicly available on Mendeley Data (DOI: 10.17632/4swmy34scp.3). As this study exclusively analyzed pre-existing, publicly available, anonymized data, no additional ethics approval was required. All analyses were conducted in accordance with relevant guidelines and regulations.

Code Availability

All source code for data preprocessing, model training, evaluation, and visualization is publicly available at: https://github.com/saivenkat-h/TH-DAT

Data Availability

Both datasets used in this study are publicly available. PERI_DEP: Zenodo, DOI: 10.5281/zenodo.11094957 (Khan et al. 2025). Uganda EPDS: Mendeley Data, DOI: 10.17632/4swmy34scp.3 (Nakalema et al. 2022).

Reporting and Reproducibility

Preprocessing: Missing values were imputed using column-wise median imputation (sklearn SimpleImputer). Categorical variables were label-encoded. Nine PHQ-9 item-level columns were excluded from the PERI_DEP feature set to prevent label leakage, yielding 19 clinical features after adding derived Trimester. The Uganda EPDS dataset used 13 clinical features.

Data Splitting: Stratified 70/15/15 train/validation/test split (random_state=42). SMOTE oversampling was applied exclusively to the training set after splitting. Validation and test sets contain only original, non-synthetic samples.

Normalization: StandardScaler fitted on training data; validation and test sets transformed using training-set statistics only.

Classification Threshold: A fixed probability threshold of 0.50 was applied consistently across all models for binary classification. This threshold was not optimized on the test set.

Hyperparameters: TH-DAT: dim=128, layers=6, heads=8, dropout=0.1, batch_size=128, 80 pretrain + 300 finetune epochs (or 200 epochs without pretraining), mixup alpha=0.2 (50% probability), label smoothing=0.05, CosineAnnealingWarmRestarts (T_0=50, T_mult=2), SWA after epoch 200, AdamW (lr=5e-4, weight_decay=1e-4). Hyperparameters were selected via validation-set performance and not tuned on the test set.

Hardware and Software: Experiments were conducted on Google Colab with NVIDIA T4 GPU (16GB VRAM). Software: Python 3.10+, PyTorch 2.x, scikit-learn 1.3+, imbalanced-learn 0.11+, XGBoost 2.0+. All random seeds fixed to 42 for primary experiments; additional seeds (123, 456, 789, 2024) used for robustness analysis.

Metrics Computation: All reported AUC-ROC, AUC-PR, F1, MCC, sensitivity, specificity, and calibration metrics were computed from actual model predictions (predict_proba for tree-based models, sigmoid outputs for neural models) on the held-out test set. ROC and PR curves visualize the discriminative performance of the top models across decision thresholds.

Model Parameters: TH-DAT total parameters: 1,693,220. All models were trained from random initialization (Xavier uniform for linear layers, standard PyTorch defaults for transformer components).

Author Contributions

S.H. conceived and designed the study, developed the TH-DAT architecture, implemented all models and baselines, conducted all experiments, performed statistical analysis, and wrote the manuscript. L., A., and S. contributed to data preprocessing, baseline implementation, and result visualization. B.N. and R.M. supervised the project, provided clinical domain expertise for feature domain mapping, reviewed the methodology and experimental design, and critically revised the manuscript for intellectual content. All authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

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